N-F-170F
March 22, 1974
SUPERSEDING
Fed. Spec. N-F-170E
April 22, 1968

FEDERAL SPECIFICATION

FEED, DRY, FOR DOGS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administraion, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for the preparation and packaging of a dry and complete feed capable of providing a suitable ration for dogs. The product may be held up to nine months under normal nonrefrigerated conditions of storage and distribution prior to consumption. (See 4.2.6.3.2 for product qualification).

1.2 Classification. The product covered by this specification shall be of the following compositions, styles of composition 1, and styles of composition 2, as specified (see 6.1):

Composition 1 - Dry
Style
A - Kibble (expanded)
B - Pellets (expanded)
C - Chunks (expanded)
D - Granules

Composition 2 - Semi-moist
Style
A - Chunks
B - Blocks
C - Shreds

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-C-110 - Cellophane (Coated and Uncoated Regenerated Cellulose Film).

FSC 8710

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L-P-378	-	Plastic Sheet and Strip Thin Gauge (Polyolefin).
UU-S-48	-	Sacks, Shipping, Paper.
PPP-B-601	-	Boxes, Wood, Cleated Plywood.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-P-704	-	Pails, Metal: (Shipping, Steel, 1 Through 12 Gallon).

Federal Standard:

FED-STD-123	-	Marking for Domestic Shipment (Civil agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the Prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handboods and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-T-21330	-	Treatment: Insect Resistant, for Paper.
MIL-L-35078	-	Loads, Unit; Preparation of Nonperishable Subsistence.
MIL-B-43666	-	Boxes, Shipping Inserts Consolidation.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-668	-	Sanitary Standards for Food Plants.
MIL-STD-1486	-	In-Transit Fumigation.

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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PUBLICATIONS

U.S. Department of Health, Education, and Welfare

Guide for the Care and Use of Laboratory Animals

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture

Visual Aids for Inspection of Metal Containers

U.S. Standards for Condition of Food Containers

Directory of Meat and Poultry Inspection Program Establishments, Circuits and Officials

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Regulations Governing the Meat Inspection of the U.S. Department of Agriculture

(Application for copies should be addressed to the information division, Consumer and Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Post Office Box 540, Benjamin Franklin Station, Washington, DC 20044.)

Association of Vitamin Chemists, Inc.

Methods of Vitamin Assay of the Association of Vitamin Chemists

(Application for copies should be addressed to John Wiley and Sons, Inc., Interscience Publishers, 605 Third Avenue, New York, NY 10016.)

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Association of American Feed Control Officials

Official Publication

(Application for copies should be addressed to the Treasurer, Association of American Feed Control Officials, Inc., Division of Agricultural Chemistry, P.O. Box 16390-A, Baton Rouge, Louisiana 70803.)

American Cereal Chemists Society

Wheat Chemistry and Technology

(Application for copies should be addressed to the American Cereal Chemists Society, 1955 University Avenue, St. Paul, MN 55104.)

National Research Council

Nutrient Requirements of Domestic Animals (Dogs)

(Application for copies should be addressed to the Printing and Publishing Office, National Academy of Sciences, 2101 Constitution Avenue, Washington, DC 20418.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials. The product shall be prepared from the ingredients specified in 3.1.1. All ingredients shall be clean, sound, and free from mold, rodent, insect or insect part(s), hair, other than that normally occurring in animal derived products as defined in the "Official Publication" of the Association of American Feed Control Officials (AAFCO), filth, paint, glass, metal, or any other evidence of contamination or off condition. In addition, animal products, milk products and fats and oils, shall not be over 6 months old at start of processing, and frozen meat shall be uncured and shall have been stored at a temperature of 0°F. or lower.

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3.1.1 Feed components. All feed components used in the formulation of the finished product shall be in accordance with the Official Listing of feed names and definitions of the Official Publication of the Association of American Feed Control Officials, hereinafter referred to as the "Official Publication".

3.2 Processing. The product shall be prepared from any combination of ingredients listed in the Official Publication in the section dealing with official and tentative definitions of feed ingredients, necessary to yield a product which will comply with 3.3. For Composition 1 product, not less than 18 percent, by weight of the total protein, shall be protein derived from animal products; for Composition 2 product, not less than 18 percent by weight of the total protein shall be derived from fresh meat or poultry, or meat, fish and poultry meats. Composition 1, styles A, B and C shall be expanded form using methods and equipment in accordance with good industry practice. Records of formulation on a percentage by weight basis shall be maintained.

3.3 Finished product. The product shall be uniform in appearance as pertains to color and texture, as applicable, and shall be free from foreign materials (e.g. charred particles) and evidence of insect infestation (e.g. insect eggs, larvae, webbing etc.) See exception in 3.1. Composition 1 and 2 products shall comply with the requirements of table I for piece size, and table II and 3.3.3 for additional nutrient requirements. Composition 2, style B (blocks) shall weigh not less than 3 ounces.

3.3.1 Acceptability requirements. Acceptability shall be determined by feeding exclusively once a day at a rate of not less than 35 calories per pound of body weight to five dogs for 21 days (see 4.3.5). The dogs shall be healthy, well-conditioned, two to six-year-old working German Shepherds weighing a minimum of sixty pounds. The dogs shall have been on full feed for at least 30 days prior to the beginning of the test, the last four days of which shall be used to gradually introduce the dogs to the test feed. Dogs shall get at least one hour of exercise per day during the entire test period. Failure of one or more dogs to maintain individual or group weight requirements in accordance with 4.3.5 (f) or evidence of intermittent diarrhea shown by two or more dogs, even if body weight is maintained, shall be considered a non-acceptance of the finished product. (NOTE: If only one test animal exhibits intermittent diarrhea while maintaining body weight it may be withdrawn from the test program only at the discretion of a qualified veterinarian and after completing an appropriate examination of the animal. A substitute dog shall then be introduced for the excluded animal).

3.3.1.1 Testure of hardness requirement (applicable to Composition 1 only). The finished product shall comply with the water absorption test as specified in 4.3.4.

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TABLE I - Sieve requirements (percent by weight) (maximum)

	Pass through US #8	Retained on US #3/8	Retained on US #1/2	Retained on US 1 inch
Composition 1				
Style A, kibble expanded	5	-	-	0
Style B, pellets expanded	5	5	0	-
Style C, chunk expanded	5	-	-	0
Style D, granules	5	0	-	-
Composition 2				
Style A 1/ chunks	5	-	-	0
Style B blocks	5	-	-	-
Style C shreds	15	-	-	0

1/ Style A - Not less than 15 pieces per 6 ounce package.

3.3.2 Nutritional adequacy. Requirements for nutritional adequacy shall be in accordance with the analytical characteristics of table II and the requirements of 3.3.3. The nutrient level shall be adjusted so that nutritional adequacy is maintained for a period of at least 9 months (see 1.1). NOTE: Nutritional adequacy for purposes of this document shall be defined as maintenance of the specified vitamin potency of the dry dog food after storage for 9 months under user's storage conditions.

TABLE II - Analytical requirements

Nutrients	Minimum value Composition 1	Minimum value Composition 2	Maximum value Composition 1	Maximum value Composition 2
	Percent by weight			
Protein	20	16.5		

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TABLE II - Analytical requirements (cont'd)

Nutrients	Minimum value Composition 1	Minimum value Composition 2	Maximum value Composition 1	Maximum value Composition 2
		Percent by weight		
Fat	5.0	4.0	-	-
Crude fiber	-	-	5.0	3.5
Ash	-	-	10.0	8.0
Moisture	-	19.0	10.0	30.0

3.3.3 Additional nutrient requirements (essential fatty acids - linoleic or arachidonic acid, minerals and vitamins). These additional nutrient components shall be at a sufficient level so that the finished product complies with the applicable nutrient requirements referenced in the publication titled, "Nutrient Requirements of Domestic Animal": of the National Academy of Sciences - National Research Council (NAS-NRC) for adult dogs. Excessive levels of added nutrients to the feed known to be toxic to dogs shall be avoided (i.e. excessive levels of vitamins, etc.).

3.4 In-transit Fumigation of Railcar Shipments of Dry Dog Food. When shipment is made by railcar between May 1 and October 31 of each year, the shipments shall be subjected to all provisions of MIL-STD-1486. Fumigation shall be conducted only by qualified personnel. (NOTE: Training for initial application of the chemical is available from the manufacturer or his representative).

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform to the applicable provisions of the Federal Food, Drug, and Cosmetic Act and Regulation Promulgated Thereunder.

3.6 Plant qualification. The product shall originate, be processed, packaged, marked and handled in plants which meet the requirements of MIL-STD-668 or in plants which are operated under the Regulations Governing Meat Inspection of the US Department of Agriculture and listed in the Directory of Meat and Poultry Inspection Program Establishments, Circuits and Officials.

3.7 Storage of manufactured feed. The product shall be packaged and packed promptly after processing and shall not be stored more than the number of days indicated below:

(a) Level A or B packing - 30 days prior to shipment to first destination.
(b) Level C packing - 30 days prior to delivery.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling. Samples of the finished product covered by this specification and samples of any materials, components, or constituents entering into the preparation of the finished product may be taken any time by the Government inspector to determine compliance with the requirements.

4.1.1 Certificate of compliance. The Government reserves the right to check test certificates of compliance to determine the validity of the certification.

4.2 Inspection.

4.2.1 General. Inspection shall be made by the procuring agency. It may be made at the site of manufacture, both during and after processing and packaging, at any suitable point in transit, and after delivery to the point of destination.

4.2.1.1 Condition inspection of containers. When inspection of containers for condition is performed by the U.S. Department of Agriculture the U.S. Standards for Conditon of Food Containers shall apply. (For applicable criteria governing inspection of containers for the Military see 4.2.6.3).

4.2.2 Factory inspection. The passing as satisfactory of any detail of processing or materials shall not relieve the supplier of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance.

4.2.3 Final inspection. Unless otherwise specified (see 6.1), final inspection shall be made after delivery to point of destination.

4.2.4 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2.4.1 Inspection records of the examination and tests shall be complete and made available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2.5 Plant qualification conditions. The product furnished under this document shall be unacceptable if not produced and stored in plants which currently meet the qualification conditions of 3.6 (see 6.6).

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4.2.6 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.6.1 Component and material inspection. In accordance with 4.2.1 components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.6.1.1 Inspection of ingredients. Conformance of ingredients to requirements contained in the "Official Publication" shall be ascertained by examination of pertinent labels, invoices, or other valid documents. Dates on invoices shall be examined to determine the age of ingredients. In addition, each ingredient shall be examined organoleptically, as necessary, to determine conformance to condition requirements of 3.1. The sample unit for examination shall be approximately 1-pound of each major ingredient (constitutes at least 1 percent of formulation) and 4 ounces of each minor ingredient, derived from a primary container. Five sample units shall be examined. In the event a lot consists of less than five containers, a sample unit shall be extracted from each container in the lot. Ingredients which do not conform to requirements of the Official Publication, or age requirement, or condition requirements, shall be unacceptable and use of such shall be cause for rejection of the lot or of the involved finished product.

4.2.6.1.2 Insect resistant treatment of sacks. Conformance to 5.2.1.1 as concerns insect resistant treatment of outer ply of sacks, shall be ascertained by examination of markings on sacks. If sacks are not marked "IRT" or the date of sack fabrication indicates the sack will be more than 75 days old at time of shipment of product, the sacks will be unacceptable and use of such shall be cause for rejection of the involved end item.

4.2.6.2 Process examination. Examination shall be made to determine compliance with the processing requirements of 3.2 and 3.6. The source and amount of protein required for Compositon 1 and 2 products shall be ascertained by multiplying the protein test results by the percentage of meat or meal in the formula, as applicable. Noncompliance with one or more of the requirements shall be cause for rejection of the involved quantity of finished product.

4.2.6.3 Examination of end item. Examination of end item shall be in accordance with tables IV through VII and 4.2.6.3.1 through 4.2.6.3.7. The examination criteria shall be as shown in table III. Acceptable quality levels (AQLs) shall be expressed as defects per hundred units for tables IV and VI, and as percent defective for tables V and VII.

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TABLE III - Sampling plans and AQLs

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major A	Major B	Minor	Total
IV	I	1 filled and sealed primary container other than cans 1/	Primary containers	-	1.5	-	6.5
V	S-4	1 filled shipping container 2/	Shipping containers	-	-	2.5	-
VI	S-2	1 pound from a bag or sack, or 1 primary package from a case	Primary containers	-	1.0	-	6.5
VII	S-3	1 filled and sealed pail	Pails	1.0	-	-	-

1/ Applicable to all primary containers including level C shipping sack.

2/ For the purpose of this examination, only 5 packages per sample unit shall be selected and examined (applicable to Composition 2 only).

TABLE IV - Examination of primary container (other than pails)

Category		Defects
Major B	Minor	
151		Open seam or seal; or tear or hole through container
	201	Tear or hole through not more than 1 ply of sack
	202	Objectionable odor
152		Nomenclature on shipping container missing, incomplete or illegible
	203	Other required marking information on shipping container missing, incomplete, or illegible
153		Not type or grade of packaging material specified
154		Not clean

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TABLE V - Examination for net weight 1/

Category	Defect
Minor	
201	Applicable to Composition 1, all styles Less than 49 lbs. 8 ounces (level A)
202	Applicable to Composition 2, styles A and C Less than 164.8 grams (level A package, chunk or shred style)
203	Applicable to Composition 2, style B Less than 71.8 grams (level A package, block style)

1/ Report results to the nearest 8 ounces for Composition 1 products and to the nearest 0.1 grams for Composition 2 products. The lot shall be rejected if sample data indicate that lot average net weight is less than specified net weight.

TABLE VI - Examination of product 1/

Category		Defect
Major B	Minor	
151		Moldy
152		Burnt
153		Not uniform in color or texture
154		Odor not typical for the product
	201	Less than 15 pieces (applicable to Composition 2, style A)

1/ Presence of foreign material (e.g., glass, metal, wood, stones, filth, insects and insect infestation (see 3.3), hair, charred particles, etc.) shall be cause for rejection of the lot. (NOTE: See exception on hair in 3.1).

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TABLE VII - Examination of leakage 1/

Category	Defect
Major A	
101	Leakage

1/ See 4.3.3.

4.2.6.3.1 Examination of piece size. Examination of piece size shall be performed in accordance with requirements of table I and procedures of 4.3.2. The sample for examination shall be a 5-pound composite of product derived from 5 primary containers, as applicable, or from a sufficient number of packages to yield a 5 pound composite sample. The sample shall be extracted from different areas of each container. Results shall be reported to the nearest whole percent. Nonconformance to piece size requirements shall be cause for rejection of the lot.

4.2.6.3.2 Examination for acceptability and nutritional adequacy. Examination shall be made to determine compliance with the acceptability and nutritional adequacy requirements of 3.3.1 and 3.3.2. Once acceptability has been determined to be satisfactory on a particular formula and with specific ingredients, subsequent lots will be acceptable as long as they remain unchanged and no deterioration of ingredients has occurred. Compliance with the nutritional adequacy requirements shall be ascertained by examination of suppliers authentic records of formulation and vitamin tests and assays of dry dog food of the same or similar composition as specified in table II and 3.3.3. One or more nonacceptance shall be considered a failure to meet the acceptability requirements and shall be cause for rejection of the lot. In addition, failure to meet the nutritional adequacy of the product as evidenced by a lack of adequate and authentic records shall be cause for disqualifying the suppliers product.

4.2.6.3.3 Examination of shipping sack (levels A and B). The filled and closed shipping sack shall be examined in accordance with the applicable criteria of UU-S-48. The following defects shall be included in the appropriate examination when empty plastic bags are required to be included in the shipping sacks: Major (1) Less than 10 bags, (2) Size of bags inadequate to carry 2 pounds of product. In addition to the defects pertinent to the sack manufacturer's identification and IRT markings, the following defects shall be applicable to those markings on the sack required to conform to MIL-STD-129: missing, incorrect or illegible, (3) Required markings on sack missing, incorrect, or illegible. This defect shall be used in lieu of marking defects contained in UU-S-48, except those pertaining to sack manufacturer's compliance. Minor: (1) Bags not positioned as specified, (2) Not heat sealed on three sides.

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4.2.6.3.4 Examination of shipping case. The filled and closed shipping case shall be examined in accordance with the criteria in the appendix of PPP-B-636. In addition, the following defects shall be included in the examintion: Major (1) - Marking missing, incorrect, or illegible; Major (2) Missing or torn plastic bag. (3) Liner missing as applicable. Minor (1) Polyethylene bag not sealed as specified. Bag or block count not as specified. For level C pack, only the major defect pertinent to marking shall apply.

4.2.6.3.5 Examination of metal pails. Examination of metal pails shall be in accordance with the examination criteria of PPP-P-704. The following defects shall be included in table of examination: Major (1) Not type, size or style specified; Major (2) Severe dent; Major (3) Dent causing sharp ridge; Minor (1) Moderate dent; NOTE: Visual Aids for Inspection of Metal Containers, (illustration of defects only) shall be used for classification of defects. (2) Bag or block count not as specified.

4.2.6.3.6 Unit loads. When loads are required to be unitized in accordance with MIL-L-35078, examination of such loads shall be in accordance with that specificaiton.

4.2.6.3.7 Examination for closure, condition, weight limitation and marking of packed boxes. When applicable, examination shall be made of the fully packed wooden boxes (see 5.3.1) for closure, condition, weight limitation and marking. The sample unit for examination shall be one filled and closed box and the sample size shall be the number of sample units indicated by inspection level S-3. Lot size shall be expressed in terms of boxes. The acceptable quality level (AQL) expressed as defects per hundred units shall be 2.5. The following defects are to be included in the examination: (1) closure insecure; (2) box damaged so as to affect product integrity; (3) Marked net weight or actual box contents exceeds weight limitation of the pertinent box specification and (4) marking, missing, incomplete or illegible.

4.2.6.4 Testing of finished product (see 6.2 and qualified testing in 6.1). The finished product shall be tested for the analytical requirements of table II and for the additional nutrient requirements of 3.3.3. Procedures for testing shall be in accordance with 4.3. The sample for testing shall be one-pound composite derived from the number of containers indicated by inspection level S-2. Lot size shall be expressed in terms of containers. Results shall be reported to the same unit or decimal specified for the pertinent requirements except that vitamins reported in Internationl Units (I.U.) shall be reported to the nearest 50 I.U. One or more test failures shall be cause for rejection of the lot. (NOTE: Non-conforming test results shall be accepted if they fall within the permitted analytical variations (PAV) as recommended in the "Official Publication" of the AAFCO).

4.2.6.4.1 Testing for texture or hardness. The finished product, as applicable, shall be tested in accordance with the requirements of 3.3.1.1. Procedures for testing shall be in accordance with 4.3.4. The sample unit for testing shall be 2 ounces of product selected from a primary container. Lot size shall be expressed in terms of the number of primary containers. The sample size shall be the number of sample units indicated by inspection level S-1. Failure of one or more of the sample units to pass the water absorption test shall be cause for rejection of the lot.

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4.2.6.5 Examination for storage of manufactured feed prior to shipment. Examination shall be made of storage time to determine compliance with the requirements of 3.7. Records of date of pack and storage time shall be maintained. Noncompliance with the referenced requirements or missing pertinent information regarding age of pack or length of storage time prior to shipment shall be cause for rejection of the involved lot.

4.2.6.6 Examination for in-transit fumigation of railcar shipments of the finished product. When applicable, examination for in-transit fumigation of railcar shipments of the finished product shall be made in accordance with the requirements of MIL-STD-1486. Appropriate records shall be maintained of manifest data, volume of the freight car, amount of fumigant used, fumigation temperatures and time, signature of qualified fumigant operators and other pertinent information. Nonconformance to one or more referenced requirements including procedures at destination reflected by actual examination or by records shall be cause for rejection of the involved quantity of finished product.

4.3 Tests. Test procedures and controls which differ from those specified herein, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.3.1 Chemical analysis. Chemical analysis shall be made in accordance with the following methods in Official Methods of Analysis of the Association of Official Analytical Chemists (AOAC). In addition, analysis for Vitamins shall be made in accordance with Methods of Vitamin Assay of the Association of Vitamin Chemist:

Test	Method
Protein 1/	Crude Protein (Kjeldahl)
Fat 1/	Crude Fat or Ether Extract
Linoleic or Arachidonic Acids 2/	Polyunsaturated Acids - Spectrophotometry
Crude Fiber 1/	Crude Fiber
Ash 1/	Ash
Moisture 1/	Moisture

Minerals 1/	Method
Calcium	Calcium
Phosphorous	Phosphorous
Potassium 3/	Methods (9) or (10)
Sodium Chloride 3/ 4/	Sodium

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Minerals 1/	Method
Magnesium	Atomic Absorption Spectrophotometry
Iron	" " "
Copper	Copper
Cobalt	Cobalt
Manganese	Atomic Absorption Spectrophotometry
Zinc	" " "
Iodine	Iodine in Mineral Mixed Feed

Vitamins 5/	Method
Vitamin A	Colorimetric
Vitamin D	Chemical
Vitamin E (alpha tocopherol)	Chemical
Vitamin B_{12}	Microbiological
Folic Acid	" "
Thiamine	Thiochrome
Riboflavin	Fluorometric
Pyridoxine	Microbiological
Pantothenic Acid	" "
Niacin	" "
Choline	Reineckate
Vitamin K	Whole Blood Prothrombin Clotting Time

Amino Acids 5/ 6/

Arginine
Histidine
Isoleucine
Leucine
Lysine
Phenylalanine
Tyrosine
Phenylalanine and Tyrosine
Methionine
Cystine
Methionine and Cystine
Threonine
Tryptophane
Valine

1/ Chapter: Animal Feed, unless otherwise specified.

2/ Chapter: Oils and Fats.

3/ Chapter: Fertilizers.

4/ Sodium chloride shall be calculated from a determination of sodium.

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5/ Test shall be conducted only when specified (see 6.1).

6/ Methods for amino acids shall be limited to ion-exchange Column Chromatography, following manual instructions supplied by amino acid analyzer instruments, such as, but not limited to, Beckman, Phoenix, etc.

(NOTE: Precautionary procedures relating to the hydrolysis of the protein to free amino acids are given in Chapter; Proteins and the Amino Acid Composition of Wheat Fractions: Section: Amino Acid Composition of Wheat Fractions in the book titled "Wheat Chemistry and Technology" published by the American Association of Cereal Chemists, Inc., St. Paul, MN 55104).

4.3.2 Sieving test for finished product. Procedure pertinent to determine piece size of the various forms of product shall be as follows:

(1) Nest the required sieves with the coarser sieve on top, and attach a collecting pan to the bottom sieve.

(2) Place a quantity of product equal to approximately 1/4 the volume of a sieve in the top sieve and attach a cover to the top sieve.

(3) Tip the sieves back and forth through a 90° arc and occasionally agitate the product by briskly moving sieves in a horizontal plane with a linear or circular motion. Sustain this procedure for three minutes.

(4) Follow the above procedures (1, 2, 3) until the five-pound sample is utilized.

(5) Weigh product retained on top sieve and passing through each individual sieve, respectively, and calculate percentage of each.

4.3.3 Leakage test. The filled and closed pail shall be tested for leakage by submerging the pail in water in such a manner that the pail is lying on its side with not less than 3 inches of water over any portion of the pail, and not less than 1-inch of water between the lower portion of the pail and the bottom of the tank. The pail shall be held in this position for not less than 30 seconds and shall then be rotated so that observation for leaks can be made over the entire circumference of the pail. The temperature of the water shall be 100° ± 5°F. A leak consists of a steady progression of bubbles. Isolated bubbles caused by entrapped air shall not be considered as a sign of leaks.

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4.3.4 Water absorption test. The product shall be rehydrated in an excess of water at a temperature of 70°F. to 80°F. for 1 hour and drained. A defective sample unit is a unit which, after draining, more than 50 percent of the pieces in the sample unit will not break up or disintegrate when subjected to pressure (by hand) from a hard object, such as the flat side of a table knife blade or a table fork. Fully moisture laden (spongy) pieces shall be acceptable.

4.3.5 Feeding regimen. The care and feeding facilities shall be in accordance with U.S. Department of Health, Education and Welfare publication "Guide for the Care and Use of Laboratory Animals".

(a) Dog shall be on full feed for at least 30 days prior to beginning of test period.

(b) Weigh dog prior to first feeding on first day of test which shall be 24 hours after the pre-test feeding. Weight of the dog shall be recorded to the nearest pound and the feed shall be weighed to the nearest ounce.

(c) Test feed shall be readily accessible to the dog for a 3-hour feeding period.

(d) Potable drinking water shall be provided separately or an ad libitum basis.

(e) Twenty-four hours after the last feeding on the 21st day of the feeding test, the dogs shall be weighed.

(f) In comparing the group average initial and group average terminal test weights, the average initial weight shall be sustained. The group average terminal weight shall be at least equal to the group average initial weight. No individual dog shall have lost more than 5 percent of its initial weight.

(g) All test and weight records including observations regarding character of the stools shall be maintained and made available for inspection.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1). The net weights and allowable unit variations shall be as follows:

Container	Net weight	Allowable unit variation
Bag	50 lbs.	- 8 ozs.
Bag	6 ozs.	- 3/16 ozs.
Bag	3 ozs.	- 3/16 ozs.

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5.1.1 Level A. The product shall be packaged in accordance with 5.1.1.1 or 5.1.1.2, as applicable.

5.1.1.1 Composition 1 product. Fifty pounds of the composition 1 product shall be packaged into a heat sealed plastic bag constructed of 0.0005-inch thick polyethylene in accordance with type I, class 2, grade B, finish 1 of L-P-378.

5.1.1.2 Composition 2 product. Six ounces of chunks or shreds shall be packaged into a heat sealed bag, or a 3-ounce block shall be packaged into a conforming, heat sealed wrap or bag. Bag and wrapping materials shall be 0.00075-inch thick cellophane complying with type VI or type VII, class 2 of L-C-110.

5.1.2 Level C. The product shall be packaged to afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A. The product shall be packaged in accordance with 5.2.1.1 or 5.2.1.2, as applicable.

5.2.1.1 Composition 1 product. A fifty pound bag of composition 1 product, packaged as specified in 5.1.1.1, shall be packed into a multiwall paper shipping sack, fabricated and closed in accordance with type I, style B, or type II, style A, construction 8 or 8 x MB1 of UU-S-48. The outer surface of the sack shall be insect-resistant treated in accordance with MIL-T-21330.

5.2.1.1.1 Individual ration bags. When specified (see 6.1), 10 empty plastic bags of sufficient size to carry 2 pounds of product, constructed of 0.00075-inch thick polyethylene in accordance with type I, class 2, grade B, finish 1 of L-P-378 and heat sealed on 3 sides, shall be placed on top of the filled and sealed polyethylene bag of product prior to closing the shipping sack.

5.2.1.2 Composition 2 product. Seventy-two 6 ounce bags of shreds or chunks or 144 wrapped blocks shall be packed in accordance with 5.2.1.2.1 or 5.2.1.2.2, as specified (see 6.1, 6.2 and 6.3).

5.2.1.2.1 Pails (see 6.2). The product shall be packed into a metal pail of 6 gallon capacity complying with type II, class 2 of PPP-P-704. The pails shall not leak when tested in accordance with 4.3.3.

5.2.1.2.2 Fiberboard box (see 6.3). The product, in a 0.005-inch thick polyethylene bag complying with type I, class 2, grade B, finish 1 of L-P-378 shall be packed into a snug-fitting fiberboard box with a liner, constructed,

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closed and reinforced in accordance with style RSC-L, V2s of PPP-B-636. The polyethylene bag shall be formed to fit the interior of the fiberboard box and after filling shall be heat sealed.

5.2.2 Level B.

5.2.2.1 Composition 1 product. Fifty pounds of Composition 1 product shall be packed into a multiwall paper shipping sack, fabricated and closed in accordance with type I, style B, or type II, style A, construction 4 or 4x of UU-S-48. The outer surface of the sack shall be insect-resistant treated in accordance with MIL-T-21330.

5.2.2.2 Composition 2 product.

5.2.2.2.1 Military agencies. Composition 2 product packed in a 0.005-inch thick polyethylene bag complying with type I, class 2, grade B, finish 1 of L-P-378 shall be packed into a snug-fitting fiberboard box with a liner constructed and closed in accordance with style RSC-L, V3c, V3s or V4s of PPP-B-636.

5.2.2.2.2 Civil agencies. Composition 2 product shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC, type CF or SF, class domestic, method II closure of PPP-B-636. The box shall have a plastic liner which is sealed or closed with a paper or plastic coated wire tie. The box shall be closed in accordance with the box specification.

5.2.3 Level C. The shipping container shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Unit loads. When specified (see 6.1) the product packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.3.1 When specified (see 6.1) the product, packed as specified in 5.2 shall be assembled into boxes conforming to PPP-B-601 or type I or II of MIL-B-43666 as specified, in quantities which do not exceed the weight limitations of the box specification.

5.4 Marking of shipping containers.

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5.4.1 Civil agencies, shipping containers. Shipping containers shall be marked in accordance with FED-STD-123 and the Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder. In addition, the marking shall include crude protein, fat, fiber, ash and moisture content, caloric value per ounce, and instructions for feeding.

5.4.2 Military agencies.

5.4.2.1 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129 and the Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder except that day, month and year of pack shall be included. In addition, the marking shall include crude protein, fat, fiber, ash and moisture content, caloric value per ounce, and instructions for feeding. The multiwall paper shipping sacks shall be marked in accordance with UU-S-48.

5.4.2.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number and date of this specification.
(b) Composition, style (Composition 1), and style (Composition 2) of product required (see 1.2).
(c) When vitamins and amino acids are to be tested.
(d) When final inspection is to be made other than after delivery to the initial point of domestic destination (see 5.1 and 5.2).
(e) Levels of packaging and packing required (see 5.1 and 5.2).
(f) When individual ration bags are required (see 5.2.1.1.1).
(g) Type and class of unit load required (see 5.3).
(h) Additional labeling and marking required (see 5.4.1.2).
(i) How unit loads are to be furnished (see 5.3 and 5.3.1).

6.2 Testing of the finished product shall be performed on the initial production lot (see footnote 5/ to 4.3.1 and 6.1). If test results meet specified and referenced requirements subsequent testing during the contract period shall be performed at a six month interval, except that testing shall be resumed in the event of any change in formulation or change in source of component feed material. The sample for testing shall be as specified in 4.2.6.4. A lot shall represent a day's production.

6.3 (Military agencies). Level A packing specified in 5.2.1.2.1 is intended for shipment to and storage in tropical areas.

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6.4 (Military agencies). Level A packing specified in 5.2.1.2.2 is intended for shipment to and storage in nontropical areas.

6.5 Destination inspection (for civil agencies only). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quantity only.

6.6 Award of contract for the product specified in this document will be limited to plants known to maintain the required sanitation conditions of 3.6 and 4.2.5.

6.7 (Military agencies). Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15, NAVSUPINST 4030.28, AFR 71-6, MCO 4030.14D or DSAR 4145.7, as applicable.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Review activities:

Army - MD
Air Force - 03

Preparing activity:

Army - GL

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
HEW-FDA, NIH
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DOCUMENT IDENTIFIER AND TITLE
Feed, Dry, for Dogs N-F-170F

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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